

COURSE OVERVIEW

Introduction to Computer and Network Security

Silvio Ranise [silvio.ranise@unitn.it or ranise@fbk.eu]



UNIVERSITÀ
DI TRENTO



COURSE INFORMATION

- Lecturer: Silvio Ranise [silvio.ranise@unitn.it or ranise@fbk.eu]

Department of Mathematics, University of Trento
&
Center for Cybersecurity, Fondazione Bruno Kessler

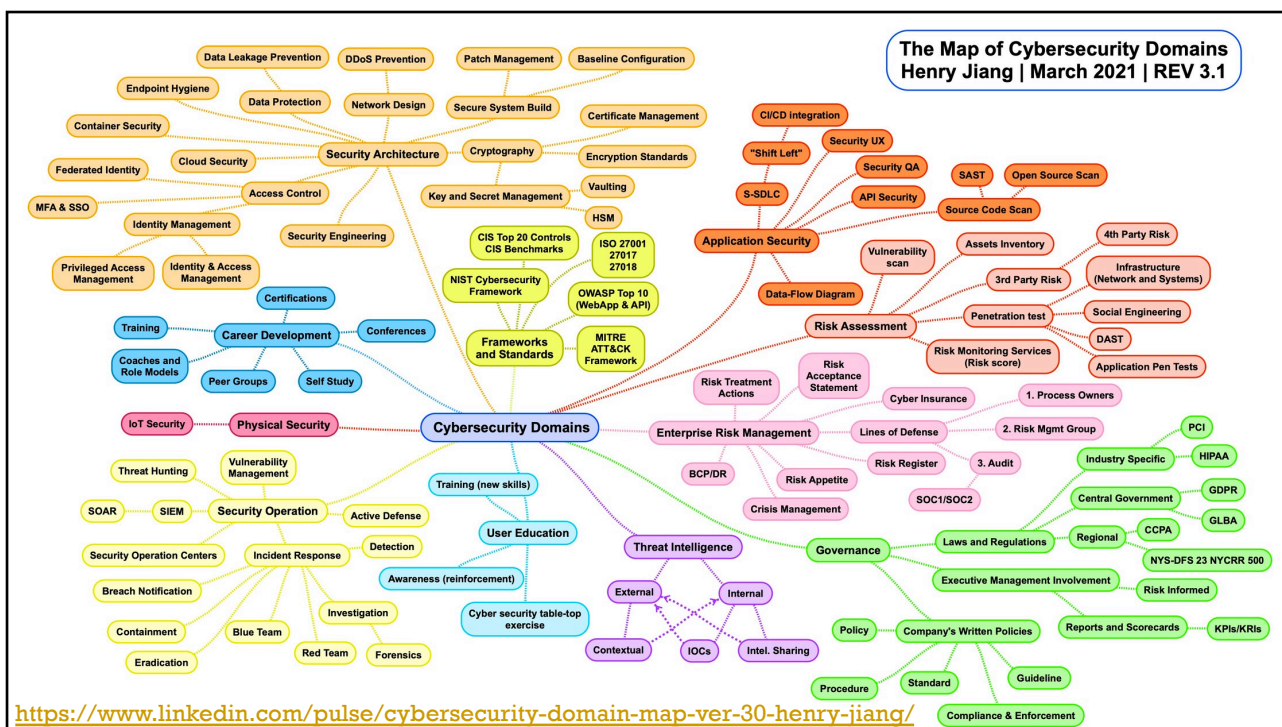
- Office hours
By appointment only. Please email me.



BE CAREFUL

- Cyber security is getting more and more import in curricula at all education levels
- Foundations of cyber security are under active development and lack a systematic view compared to other disciplines (such as Mathematics, ...)
- Efforts are ongoing to develop a coherent view of the discipline
 - Example: <https://cybered.acm.org/>
- However, there is not even a single, agreed-upon, definition of cyber security
 - <https://www.enisa.europa.eu/publications/definition-of-cybersecurity>
- Keep in mind that the term “cybersecurity” has several different ramifications
 - Information security substantially contributes to cyber security
 - Computer and Network security can be seen as part of cyber security
- Cyber security is conceptually spread over **several different domains**

2



This course...

AIMS & OBJECTIVES

By the end of the course, you should be able to:

- understand the **theoretical** and **practical** problems of **information security**
- understand and recognize **threats** to fundamental security properties: **confidentiality, integrity, and availability**
- understand how the main **security mechanisms** - such as **authentication** and **authorization** together with **access control** and **cryptography** - can be applied to **mitigate vulnerabilities**

TOPICS

- Basic notions
- Authentication
- Introduction to cryptography
- Applications of cryptography
- Access Control
- Application security
- Privacy & GDPR

Focus on how to **defend** (more precisely, how to mitigate risks) rather than how to **attack** (although some basic ideas on how to exploit vulnerabilities to perform attacks will be given)

4

EXAM

- Written exam (questions and/or exercises) at the end of the course
- Dates: TBA
- Past papers and results of past exam sessions are available online

<https://sites.google.com/view/intro2cns/>

5

TEACHING

- I will use mainly slides but occasionally will suggest some reading material that can be found online
- Lectures are recorded (Wednesday 16:30-18:30 and Thursday 13:30-15:30)
- Materials will be made available on Moodle

- You can find some useful material in the following book
 - *Computer Security*
A book by Dieter Gollmann
- And also at the following site
 - <https://www.cybok.org/>



PROJECTS

Opportunities for curricular internship and/or thesis may arise, subject to approval from the lecturer

- **Workload:** approximately one week of full-time work per academic credit, including reading, writing, coding, preparation, and presentation – e.g., 6 weeks for undergraduate internship
- **Location:** *Center for Cyberscurity* (FBK north building, first floor)
- Past project topics are available online (FYI)
 - <https://st.fbk.eu/teaching/projects/2020-2021>
- This year's topic list is **not yet available**
- Interviews are **not open yet**

